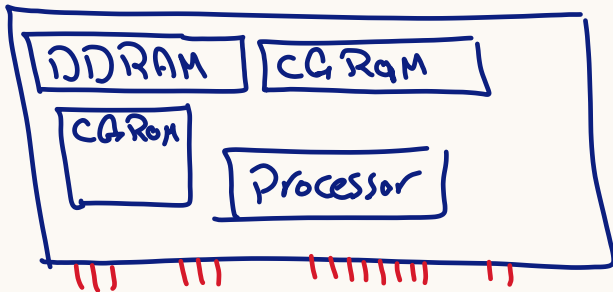
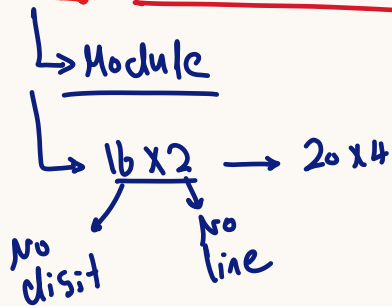


LCD: Liquid crystal display



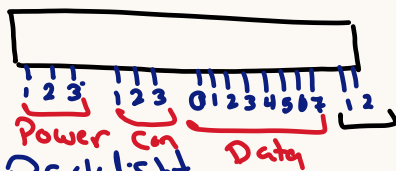
Pinout:

Power

- ① Gnd → VSS
- ② VCC → VDD
- ③ V₀ → Pot

Data Pin

- ① character → ASCII
- ② Data length
 - 1 byte → 8 bit at one time
 - 1 byte → 4 bit at twice (D4 → D7)



Back light

- BLA: Anode → VCC
BLK: Cathode → Gnd.

Control Pin

① RS: Register select



② RW: Read/Write

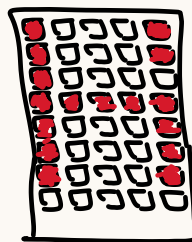
- RW → 1 → Read
RW → 0 → Write

③ E: Enable

- E → High
E → Low

How draw and character

* Disits → 5x8 → Pixel
5x11



Processor

→ Data Display

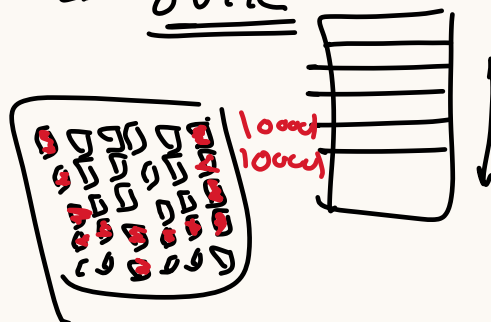
→ CGROM
→ DDRAM

CGROM: character generator ROM

→ Memory store all want to draw all characters

CGRAM: Memory store for special character

→ 8 location → 8 char 00
→ 8 bits 5x8



DDRAM: Data Display

① Power

- ① Gnd → Gnd.
 - ② VDD → 5V
 - ③ V₀ → Pot
- ② Control: 1. RS → Mc
RW → Mc E → Mc

③ Data Pin

- 8 bit Mode
D0 → D7
Mc
- 4 bit Mode
D4 → D7

④ Back light

- A → VCC K → Gnd.

00	0F
0 Trut	15 Hexam
0	15
40	4F

line 1 Address → 0x00

line 2 Address → 0x40

void LCD_SendInstruction
(uint8_t Command)

- ① RS → 0
- ② RW → 0
- ③ Data → Command
- ④ En → H
- ⑤ wait(1 / μs)
- ⑥ En → low

void LCD_WriteChar
(uint8_t character)

- ① RS → 1
- ② RW → 0
- ③ Data = character
- ④ En → H
- ⑤ wait → 1 μs
- ⑥ low → L

CGRAM →

$$\text{Address} = \text{Location} \times 8 \rightarrow \underline{0x08}$$

$$= 00001000$$

① A₅ A₄ A₃ A₂ A₁ A₀

Address | 0 0 0 0 1 0 0 0

set CGRAM | 0 1 0 0 0 0 0 0

Address	0	0	0	0	1	0	0	0
set CGRAM	0	1	0	0	0	0	0	0
	0	1	0	0	1	0	0	0

← set Bit (Address, 6);

lcd_SendCommand(0x40 | Address);

$$\begin{array}{r} 0A \\ \hline 0x80 | 0x0A \end{array}$$